

THE RHOMBICOSIDODECAHEDRON IS NOT RUPERT

GIHYO JUNG

ABSTRACT. A convex body is *Rupert* if a hole can be cut through it such that a congruent copy can pass through the hole. Steininger and Yurkevich recently constructed the first convex polyhedron without this property; earlier they had conjectured that the rhombicosidodecahedron is not Rupert either. We prove this conjecture by a computer-assisted proof. The subdivision method of Steininger and Yurkevich does not suffice for the rhombicosidodecahedron on its own: because of its large symmetry group there are one-parameter families of configurations in which the two shadows coincide, and at four exceptional configurations the first-order protection that the method relies on degenerates. We describe these coincidences, locate the four singular configurations exactly in $\mathbb{Q}(\sqrt{5})$, and prove three new local certificates: a fixed-direction coincidence theorem, a multi-contact gradient theorem and a blow-up (sector and arc-tube) theorem at singular points. All inequalities of the certificates are verified in outward-rounded interval arithmetic, and all structural identities are verified in exact arithmetic in $\mathbb{Q}(\sqrt{5})$.

1. INTRODUCTION

According to John Wallis (1693), Prince Rupert of the Rhine wagered that a hole can be cut in a cube through which a cube of the same size can pass. A convex body $K \subset \mathbb{R}^3$ is called *Rupert* if there are rotations $O_1, O_2 \in SO(3)$ and a vector $t \in \mathbb{R}^2$ such that

$$(1) \quad \Pi O_1 K + t \subset \text{int}(\Pi O_2 K),$$

where $\Pi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is the orthogonal projection $(x, y, z) \mapsto (x, y)$. All five Platonic solids and many Archimedean and Catalan solids are Rupert [5, 1, 4, 6, 2]. Steininger and Yurkevich [8] constructed the first convex polyhedron that is not Rupert (the *Noperthedron*). Earlier, in [7], they proved that the truncated icosidodecahedron is Rupert, so that ten of the thirteen Archimedean solids are known to be Rupert, and conjectured on statistical evidence that the rhombicosidodecahedron is not [7, Conjecture 2]; the snub cube and the snub dodecahedron remain open. Extensive numerical searches [7, 3] found no passage for the rhombicosidodecahedron. (For computational evidence that a certain stellated tetrahedron is not Rupert, see [9].)

Theorem 1.1. *The rhombicosidodecahedron is not Rupert.*

Method and difficulties. Steininger and Yurkevich [8] parametrise the pairs of projections by a five-dimensional box, subdivide it, and exclude every sub-box either by a *global theorem* (a separating direction that works on the whole box) or by a *local theorem* near configurations in which the two shadows coincide. For the Noperthedron, which was designed to have a small symmetry group, these two tools suffice. In [8, Section 9.1] the authors explain that for the rhombicosidodecahedron their local theorem cannot be applied at the view along a coordinate axis of the standard coordinates (a two-fold axis; this is our point A below) and in about 2% of random view directions. For the rhombicosidodecahedron $\mathcal{P}$ (symmetry group I_h of order 120) they do not, for three reasons which we identify in Section 3:

- (i) there are many more coincidences: the shadows coincide whenever $O_1 = D O_2 g$ with g in the rotation group I and $D \in \{I, \rho_\pi\}$ (ρ_π the half-turn about the viewing axis); in the reduced parameter domain two such branches cross at views along a two-fold axis; and on

- an arc of *half-twist* configurations, in which the inner body is the outer one rotated by 36° about a five-fold axis, the inner shadow touches the outer one from inside;
- (ii) at a view along a two-fold axis and at a view orthogonal to a five-fold axis (points A, B) the linear protection of the local theorem degenerates: the protrusion of the inner shadow is only of order $|w| |\xi|$, where w is the offset of the view and ξ the relative rotation;
 - (iii) at the two endpoints C, D of the half-twist arc the protection degenerates in a different way (the protrusion vanishes identically along a curve).

Near the coincidences of (i), boxes are closed by the local theorem of [8], whose orthogonal map need not be a symmetry of the polyhedron (this closes the interior of the half-twist arc), and by a fixed-direction version of the local theorem (Theorem 4.1); a multi-contact gradient certificate (Theorem 4.2) closes boxes close to, but not containing, coincidences. We deal with (ii), (iii) by blow-up certificates (Theorems 4.6 and 4.9) that are valid in a whole neighbourhood of the singular configuration, including the singular configuration itself.

Rigour. All inequalities of the certificates are checked in outward-rounded interval arithmetic (Section 6.1); the few combinatorial decisions taken in floating point, all with large slack, are listed in Appendix C. All statements of the form “this quantity is exactly zero” (collinearity of projected vertices at a singular view, identical vertex trajectories, exactness of a contact along the half-twist arc, a vertex lying in the shadow plane) are proved in exact arithmetic in $\mathbb{Q}(\sqrt{5})$, using exact coordinates of the singular configurations (Proposition 3.1). The code and the certificates are available at <https://github.com/swiri0921-bit/rhombicosidodecahedron-not-rupert> (release v1.0; the file `results/CODE_HISTORY.md` lists the hashes of the programs used for the runs).

Use of AI. The author chose the problem, planned and directed the project, and made the decisions on how to proceed at each stage. In particular, the author proposed to write out and audit the hand proofs of all certificates while the computations were running; this audit led to the detection and correction of several errors before the final run. The author ran a large part of the computations on his own computer, monitored them to completion, and prepared the public repository. An AI system (Claude, by Anthropic) was used extensively as a research assistant: for the literature search, for developing the mathematical arguments and the programs, for running computations, and for drafting the text; the second and third implementations used for the checks of Section 6.2 were written with the same assistance. The proof does not ask the reader to trust the AI system: all statements are proved in this paper, and every computer-verified statement can be reproduced and checked with the published programs and result files.

2. SETUP

2.1. The polyhedron. Let $\varphi = (1 + \sqrt{5})/2$. The rhombicosidodecahedron $\mathcal{P}$ is the convex hull of the 60 points obtained from

$$(\pm 1, \pm 1, \pm \varphi^3), \quad (\pm \varphi^2, \pm \varphi, \pm 2\varphi), \quad (\pm(2 + \varphi), 0, \pm \varphi^2)$$

by cyclic permutations of the coordinates. We rotate $\mathcal{P}$ so that the five-fold axis through $(0, \varphi, 1)$ becomes the z -axis and scale it to circumradius 1; we denote its vertices by $V_1, \dots, V_{60}$, $|V_k| = 1$. The rotation is not rational; wherever exact values are needed we work in the body frame (coordinates in $\mathbb{Q}(\sqrt{5})$), and the rotated coordinates are enclosed in intervals computed from decimal values with error below 10^{-55} (Section 6.1). $\mathcal{P}$ is centrally symmetric: $-V_k = V_{k'}$.

2.2. Parametrisation. Following [8], for $\theta, \phi \in \mathbb{R}$ let

$$M(\theta, \phi) = \begin{pmatrix} -\sin \theta & \cos \theta & 0 \\ -\cos \theta \cos \phi & -\sin \theta \cos \phi & \sin \phi \end{pmatrix}, \quad X(\theta, \phi) = (\cos \theta \sin \phi, \sin \theta \sin \phi, \cos \phi),$$

so that $M(\theta, \phi)$ is the projection along the view direction $X(\theta, \phi)$, and let $R(\alpha)$ be the rotation of $\mathbb{R}^2$ by α . We write $A(\theta, \phi) \in SO(3)$ for the matrix with rows the two rows of M and X . Every $O \in SO(3)$ satisfies $\Pi O = R(\alpha)M(\theta, \phi)$ for suitable (θ, ϕ, α) .

Lemma 2.1 (reductions). *If $\mathcal{P}$ is Rupert, then there is $x = (\theta_1, \phi_1, \theta_2, \phi_2, \alpha)$ in*

$$\mathcal{D} = [0, 2\pi/5] \times [0, \phi_{\max}] \times [0, 2\pi/5] \times [0, \phi_{\max}] \times [-\pi/2, \pi/2], \quad \phi_{\max} = 0.6524,$$

with $R(\alpha)M(\theta_1, \phi_1)\mathcal{P} \subset \text{int } M(\theta_2, \phi_2)\mathcal{P}$ (no translation).

Proof. Translation. If $S + t \subset \text{int } T$ for centrally symmetric convex sets S, T , then also $S - t = -(S + t) \subset \text{int } T$, and by convexity $S \subset \text{int } T$ (see [8]). *Symmetries.* Replacing O_i by $O_i g_i$ with g_i in the rotation group $I \subset I_h$ does not change the bodies $O_i \mathcal{P}$. The view direction of O_i is $O_i^T e_3$, and every unit vector is within the angle $\arccos \sqrt{(5 + 2\sqrt{5})/15} = 0.652358 \dots < \phi_{\max}$ (about 37.38°) of one of the twelve five-fold axes (this is the angle between a five-fold and an adjacent three-fold axis, the covering radius of the five-fold axes), so we may assume that both view directions are within $\phi_{\max}$ of e_3 . Since $M(\theta + 2\pi/5, \phi) = M(\theta, \phi)R_z(-2\pi/5)$ and $R_z(2\pi/5) \in I$, $\theta_i \in [0, 2\pi/5]$. Finally $R(\alpha + \pi) = -R(\alpha)$ and $\mathcal{P} = -\mathcal{P}$, so $\alpha \in [-\pi/2, \pi/2]$. The outer rotation about the view axis can be absorbed into α . $\square$

We call $x \in \mathcal{D}$ a *passage* if $R(\alpha)M(\theta_1, \phi_1)\mathcal{P} \subset \text{int } M(\theta_2, \phi_2)\mathcal{P}$. Theorem 1.1 is equivalent to: $\mathcal{D}$ contains no passage. To exclude a set of configurations it suffices to exhibit, for each x , a direction $n \in \mathbb{R}^2 \setminus \{0\}$ and an inner vertex i with

$$(2) \quad P_{i,n}(x) := \langle p_i(x), n \rangle - \max_k \langle q_k(x), n \rangle \geq 0, \quad p_i = R(\alpha)M(\theta_1, \phi_1)V_i, \quad q_k = M(\theta_2, \phi_2)V_k.$$

Indeed, $P_{i,n} \geq 0$ says that p_i is not in the interior of the outer shadow.

2.3. Derivative bounds. All maps below are products of rotations about fixed axes, each depending on one parameter at unit speed.

Lemma 2.2. *Let $R_3(\theta)$ be the rotation about e_3 by θ . Then $A(\theta, \phi) = B(\phi)R_3(\theta)^T$, where the rows of $B(\phi)$ are e_2 , $-\cos \phi e_1 + \sin \phi e_3$ and $\sin \phi e_1 + \cos \phi e_3$. Consequently*

$$\partial_\theta A = A S_\theta, \quad \partial_\phi A = S_\phi A, \quad \partial_\alpha \text{diag}(R(\alpha), 1) = S_\alpha \text{diag}(R(\alpha), 1)$$

with constant skew matrices $S_\theta, S_\phi, S_\alpha$ of operator norm 1. Hence all first and second partial derivatives of A , of $A_{\text{in}} = \text{diag}(R(\alpha), 1)A(\theta_1, \phi_1)$ and of their products with fixed rotations have operator norm at most 1, and:

- (a) $\|A(\theta, \phi) - A(\theta_0, \phi_0)\| \leq |\theta - \theta_0| + |\phi - \phi_0|$;
- (b) *the rotation angles of $A_{\text{in}}(x)A_{\text{in}}(x_0)^T$ and of $A_{\text{out}}(x)A_{\text{out}}(x_0)^T$ are at most $\sum_{d \in \{1,2,5\}} |x_d - x_d^0|$ and $|x_3 - x_3^0| + |x_4 - x_4^0|$, respectively;*
- (c) *for $|V| \leq 1$ and $n \in \mathbb{R}^2$ the functions $x \mapsto \langle \Pi A_{\text{in}}(x)V, n \rangle$, $\langle \Pi A_{\text{out}}(x)V, n \rangle$ differ from their first-order Taylor polynomials at x_0 by at most $\frac{1}{2}(\sum_d |x_d - x_d^0|)^2 |n|$, where d ranges over $\{1, 2, 5\}$ for the inner and over $\{3, 4\}$ for the outer body.*

Proof. The factorisation is checked directly from the definition of M and X . Since $\partial_\theta R_3(\theta)^T = R_3(\theta)^T [e_3]^T$, we get $\partial_\theta A = A [e_3]^T$, a constant skew matrix of norm 1; $B'(\phi)$ rotates the second and third rows into each other, $B' = S_\phi B$. Second derivatives are products of a rotation and two such generators, and $\|S^2\| \leq 1$. (a) and (c) follow from the mean value theorem and Taylor's theorem. For (b), the angle of $Q \in SO(3)$ is its distance to I in the bi-invariant metric, and $t \mapsto A_{\text{in}}(x_0 + t(x - x_0))$ has angular speed at most $\sum_d |x_d - x_d^0|$. $\square$

3. COINCIDENCES AND SINGULAR CONFIGURATIONS

3.1. Coincidences. A configuration x is a *coincidence* if the two shadows are equal. Coincidences are never passages, but no box containing a coincidence can be excluded by a separating-direction argument, so every coincidence must be dealt with by a local theorem. Writing the inner orientation as $A_{\text{in}} = e^{[\xi]} D A_{\text{out}} h$ with $\xi \in \mathbb{R}^3$ (rotation vector, $[\xi]$ the cross-product matrix), the following configurations have to be treated as coincidences.

- (a) *Symmetry coincidences:* $\xi = 0$, $h \in I$ and $D \in \{I, \rho_\pi\}$, $\rho_\pi = \text{diag}(-1, -1, 1)$ (rotation by π about the viewing axis; $\Pi\rho_\pi = -\Pi$ and $\mathcal{P} = -\mathcal{P}$). In the reduced domain $\mathcal{D}$ these form several branches. Two of them meet at views along a two-fold axis a : if $h_a \in I$ is the half-turn about a and $X = a$, then $A_{\text{out}} h_a A_{\text{out}}^T = \rho_\pi$, so the branches $A_{\text{in}} = A_{\text{out}}$ and $A_{\text{in}} = \rho_\pi A_{\text{out}} h_a$ coincide there, while for views near a they differ by a small rotation (Lemma A.2).
- (b) *Half-twist configurations:* $A_{\text{in}} = A_{\text{out}} T$ with T the rotation by 36° about the five-fold axis $a = (\varphi, 1, 0)$ (body frame) composed with an element of I (the exact matrices T_C, T_D are given in the repository). The axis a and the five-fold axis $w_5 = (0, \varphi, 1)$ of the viewing cap span a mirror plane and enclose the angle $\arctan 2$. The twenty vertices of the two decagonal rings orthogonal to a are mapped to vertices, and for outer views X in this mirror plane, between w_5 and a , with $\angle(X, w_5) \in [\arctan \varphi^{-5}, \arctan \frac{1}{3}]$ (so that $\angle(X, a) = \arctan 2 - \angle(X, w_5)$ lies between 45° and about 58.3°) the inner shadow lies in the outer one and shares boundary points with it (Figure 2). On this *half-twist arc* the protrusion is exactly 0 (so these configurations are not passages); the arc is not a family of coincidences in the strict sense, but it has to be treated like one. Near interior points of the arc, three inner and three outer vertices of the relevant contacts are congruent via an orthogonal map which is not a symmetry of $\mathcal{P}$; boxes there are closed by the local theorem of [8] (Section 4).

This description guides the choice of certificates, but the proof does not depend on its completeness: every box of $\mathcal{D}$ is closed by one of the certificates of Section 4.

3.2. Singular configurations. Near a symmetry coincidence x_0 with outer view X_0 the protrusion is, to first order, $\max_c \langle \xi, g_c \rangle$ where $g_c = U_i \times N$ ranges over the contacts (Theorem 4.1). If the cone spanned by the g_c is all of $\mathbb{R}^3$, a whole neighbourhood of x_0 is excluded. This fails exactly at views where several faces are seen edge-on. A numerical scan of the cap finds six views with at least two edge-on faces; at four of them the protection vanishes (Table 1). (The scan is not part of the proof; its role is to locate the points where blow-up certificates are needed.)

Proposition 3.1 (exact singular configurations). *In the body frame (coordinates as in Section 2, up to scale) the singular views are*

$$X_A = (1, \varphi, \varphi - 1), \quad X_B = (1, \frac{5+\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2}), \quad X_C = (1, \frac{9+5\sqrt{5}}{2}, \frac{5+3\sqrt{5}}{2}), \quad X_D = (1, \frac{3\sqrt{5}-1}{2}, \frac{5-\sqrt{5}}{2}).$$

The half-twist rotations T_C, T_D have entries in $\mathbb{Q}(\sqrt{5})$ in the body frame. (Geometrically, X_A is a two-fold axis, X_B is orthogonal to a five-fold axis in a mirror plane, and X_C, X_D are the endpoints of the half-twist arc; these descriptions, checked numerically, are for orientation only: the proof uses only the exact coordinates.)

Proof. The coordinates are exact elements of $\mathbb{Q}(\sqrt{5})^3$; all identities used in the certificates are verified in exact arithmetic (file `exact_check.py`, Section 6.3). For T : the rotation by 36° about $a = (\varphi, 1, 0)$ is $\cos 36^\circ I + \sin 36^\circ [\hat{a}] + (1 - \cos 36^\circ) \hat{a} \hat{a}^T$ with $\cos 36^\circ = \varphi/2$ and $\sin 36^\circ/|a| = (\sqrt{5}-1)/4$, so all entries lie in $\mathbb{Q}(\sqrt{5})$. $\square$

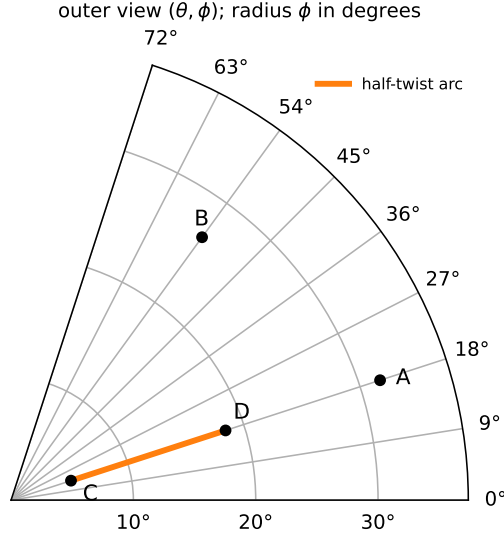


FIGURE 1. The reduced range of outer views ($\theta \in [0, 72^\circ]$, $\phi \leq 37.38^\circ$) with the singular views A, B, C, D and the half-twist arc from C ($\phi = \arctan \varphi^{-5}$) to D ($\phi = \arctan \frac{1}{3}$).

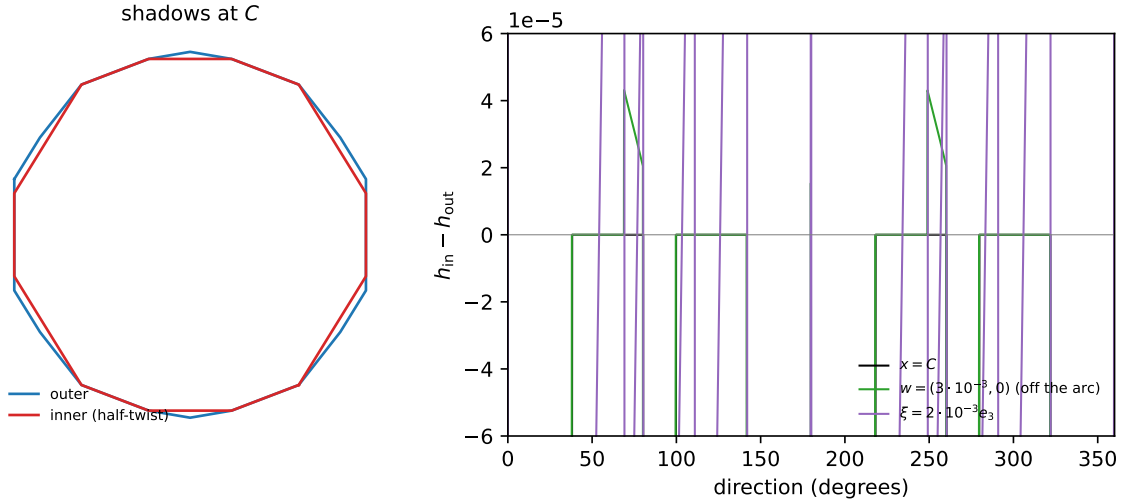


FIGURE 2. Left: outer shadow and half-twisted inner shadow at C . Right: $h_{\text{in}} - h_{\text{out}}$ (difference of support functions) as a function of the direction; at C its maximum is exactly 0 on whole intervals of directions, off the arc or after a small rotation it becomes positive. The vertical scale is 10^{-5} .

4. CERTIFICATES

We use six certificates. Each takes a box $\mathcal{B} \subset \mathcal{D}$ (or a region in blow-up coordinates) and some floating-point data (a direction, multipliers of a linear program), and each has a conclusion of the form “ $\mathcal{B}$ contains no passage” whose hypotheses are finitely many inequalities verified in interval arithmetic and finitely many identities verified in $\mathbb{Q}(\sqrt{5})$. Floating-point data are only *witnesses*: they are never assumed to be exact.

point	view	coincidence type	degeneracy	treatment
A	two-fold axis	two symmetry branches	$P \sim c w \xi $	sector, two references
B	$\perp$ five-fold axis	symmetry	$P \sim c w \xi $	sector
C	arc endpoint	half-twist (two branches)	$P \equiv 0$ on the arc	two arc tubes + sector (s -mode)
D	arc endpoint	half-twist	$P \equiv 0$ on the arc	arc tube + sector (s -mode)

TABLE 1. The four singular configurations.

4.1. The theorems of Steininger and Yurkevich. We use the global theorem [8, Theorem 17] and the local theorem [8, Theorem 36] in the form stated there (with $\rho = 1$ and ε the largest half-width of the box, rounded upward; the theorems then apply to a cube containing the box). The local theorem requires two triples of vertices to be *congruent*, i.e. related by an orthonormal matrix L [8, Definition 34]; L need not be a symmetry of the polyhedron, and near the interior of the half-twist arc it is not. We certify congruence either by a symmetry of $\mathcal{P}$ mapping one triple to the other, or by checking that the Gram matrices of the two triples are equal in $\mathbb{Q}(\sqrt{5})$, which is equivalent to the existence of such an L (possibly with determinant -1). The inequalities checked for the local theorem are listed in Appendix B.

4.2. Fixed-direction coincidence theorem.

Theorem 4.1. *Let $\mathcal{B}$ be a box with centre x_0 and largest half-width ε . Let $h \in I$ and $D \in \{I, \rho_\pi\}$, and put $U_i(x) = DA_{\text{out}}(x)hV_i$ and $e^{[\xi(x)]} = A_{\text{in}}(x)(DA_{\text{out}}(x)h)^T$, so that the inner body is $\{e^{[\xi]}U_i\}$ and $\Pi U_i = \Pi A_{\text{out}}V_{k(i)}$ for a bijection k . Let $n_1, \dots, n_m$ be unit vectors in $\mathbb{R}^2$ and k_c vertices such that for every c and every $x \in \mathcal{B}$, k_c is the unique maximiser of $\langle \Pi A_{\text{out}}(x)V_j, n_c \rangle$. Let $g_c = U_{i_c}(x_0) \times N_c$ with $N_c = (n_c, 0)$ and $i_c = k^{-1}(k_c)$, and*

$$c_0 = \min_{|v|=1} \max_c \langle v, g_c \rangle, \quad \xi_{\max} = |\xi(x_0)| + 5\varepsilon.$$

If $c_0 - 2\varepsilon > \xi_{\max}/2 + \xi_{\max}^2/6$, then $\mathcal{B}$ contains no passage.

Proof. Fix $x \in \mathcal{B}$ and write $\xi = \xi(x)$, $U_i = U_i(x)$. If $\xi = 0$ the shadows coincide and x is not a passage. Otherwise, since k_c is the unique maximiser and $\Pi U_{i_c} = \Pi A_{\text{out}}V_{k_c}$,

$$P_{i_c, n_c}(x) = \langle \Pi(e^{[\xi]} - I)U_{i_c}, n_c \rangle = \langle (e^{[\xi]} - I)U_{i_c}, N_c \rangle.$$

For a rotation by angle $\vartheta = |\xi|$ one has $\|e^{[\xi]} - I - [\xi]\| = \|(1 - \cos \vartheta)[\hat{\xi}]^2 + (\sin \vartheta - \vartheta)[\hat{\xi}]\| \leq \vartheta^2/2 + \vartheta^3/6$, hence, with $|U_i| = 1$, $P_{i_c, n_c} \geq \langle \xi, U_{i_c} \times N_c \rangle - |\xi|^2/2 - |\xi|^3/6$. By Lemma 2.2, $|U_i(x) - U_i(x_0)| \leq 2\varepsilon$, so $\langle \xi, U_{i_c}(x) \times N_c \rangle \geq |\xi|(\langle \hat{\xi}, g_c \rangle - 2\varepsilon)$, and the angle of $e^{[\xi(x)]}$ is at most $|\xi(x_0)| + 3\varepsilon + 2\varepsilon$. Choosing c with $\langle \hat{\xi}, g_c \rangle \geq c_0$ gives $P_{i_c, n_c}(x) \geq |\xi|(c_0 - 2\varepsilon - \xi_{\max}/2 - \xi_{\max}^2/6) > 0$. $\square$

Verification. The directions n_c are interior directions of the normal cones of the outer shadow at x_0 ; unique maximality on $\mathcal{B}$ is checked by $\langle \Pi A_{\text{out}}(x_0)(V_{k_c} - V_j), n_c \rangle > 2\varepsilon|V_{k_c} - V_j|$ for all $j \neq k_c$ (Lemma 2.2). The rotation angle of $e^{[\xi(x_0)]}$ is bounded from $y = \|e^{[\xi]} - I\|_F/\sqrt{8} = \sin(\vartheta/2)$ by $\vartheta \leq 2y/\sqrt{1-y^2}$. The number c_0 is bounded below by covering the sphere by caps of radius $r = 1.001 \cdot \sqrt{2}/48$ around the normalised centres of the cells of a 48×48 grid on each face of the cube $[-1, 1]^3$ (radial projection onto the sphere is 1-Lipschitz on the cube surface) and using $\langle v, g \rangle \geq \langle p, g \rangle - r|g|$. The symmetry h is enclosed from 60-digit data of the exact matrix.

4.3. Multi-contact gradient certificate.

Theorem 4.2. *Let $\mathcal{B}$ be a box with centre x_0 and half-widths $\delta^{\max} \in \mathbb{R}^5$. Let $(i_c, n_c, K_c)_{c=1}^m$ be contacts (inner vertex, direction, set of outer vertices) and $\lambda \in \mathbb{R}_{\geq 0}^m \setminus \{0\}$. Suppose*

- (a) for every c , every $j \notin K_c$: $\sup_{\mathcal{B}} \langle q_j, n_c \rangle \leq \max_{k \in K_c} \inf_{\mathcal{B}} \langle q_k, n_c \rangle$ (bounds from first-order Taylor expansion plus the remainder of Lemma 2.2);
 (b) at each of the 32 vertices δ of $\prod_d [-\delta_d^{\max}, \delta_d^{\max}]$,

$$\sum_c \lambda_c \left(\ell_{i_c}(\delta) - \max_{k \in K_c} \ell_k(\delta) - \frac{1}{2} (h_{\text{in}}^2 + h_{\text{out}}^2) |n_c| \right) > 0,$$

where ℓ_i, ℓ_k are the first-order Taylor polynomials at x_0 of $\langle p_i, n_c \rangle$ and $\langle q_k, n_c \rangle$, and $h_{\text{in}} = \delta_1^{\max} + \delta_2^{\max} + \delta_5^{\max}$, $h_{\text{out}} = \delta_3^{\max} + \delta_4^{\max}$.

Then $\mathcal{B}$ contains no passage.

Proof. By (a) the maximum in (2) is attained in K_c . By Lemma 2.2, $P_{i_c, n_c}(x_0 + \delta) \geq F_c(\delta) := \ell_{i_c}(\delta) - \max_{k \in K_c} \ell_k(\delta) - \frac{1}{2} (h_{\text{in}}^2 + h_{\text{out}}^2) |n_c|$. The function $\sum_c \lambda_c F_c$ is concave (a sum of minima of affine functions), so its minimum over the box is attained at a vertex, where it is positive by (b). Hence some $P_{i_c, n_c}(x) > 0$ for every $x \in \mathcal{B}$. $\square$

4.4. Blow-up coordinates at a singular point. Let x^* be one of A, B, C, D with outer view (θ^*, ϕ^*) , and fix a reference $A_{\text{ref}}(w) = M_f A(w) T$ for the inner body, where $A(w) = A(\theta^* + w_1, \phi^* + w_2)$, T is a fixed rotation ($T \in I$ at A, B : $T = I$, and $T = h$ for the second reference $\rho_\pi A(w) h$ at A ; a half-twist at C, D), and $M_f \in \{I, \rho_\pi\}$. Every configuration near x^* is written as

$$A_{\text{out}} = A(w), \quad A_{\text{in}} = e^{[\xi]} A_{\text{ref}}(w), \quad (w, \xi) \in \mathbb{R}^2 \times \mathbb{R}^3.$$

Put $U_i(w) = A_{\text{ref}}(w) V_i$, $W_k(w) = A(w) V_k$ and $s = \max(|w|, |\xi|)$; for $s > 0$ write $w = s\hat{w}$, $\xi = s\tau\omega$ with $|\omega| = 1$, $\tau \in [0, 1]$. Since $\Pi\rho_\pi = -\Pi$ we have $\Pi(W_k - U_i)(w) = \Pi A(w) d_{ik}$ with the fixed vector $d_{ik} = V_k - \sigma T V_i$, $\sigma = \pm 1$ according to M_f .

For a line j of the outer shadow at x^* (the projection of an edge, or a supporting line at a vertex) with outer unit normal n_j and level H_j let $K_j = \{k : \langle \Pi W_k(0), n_j \rangle = H_j\}$ and $I_j = \{i : \langle \Pi U_i(0), n_j \rangle = H_j\}$; these sets are certified exactly (Section 6.3). A *contact* is $c = (i, j, \phi, \chi)$ with $i \in I_j$ and direction $n_c = R(s\phi + \chi w_1) n_j$; $N_c = (n_c, 0)$. Its protrusion is $P_c = \langle \Pi e^{[\xi]} U_i, n_c \rangle - \max_k \langle \Pi W_k, n_c \rangle$.

Lemma 4.3 (second-order rotation formula). *For $\xi = \vartheta\omega$ ($|\omega| = 1$) and vectors U, N ,*

$$\langle e^{[\xi]} U, N \rangle = \langle U, N \rangle + \langle \xi, U \times N \rangle + \frac{1}{2} |\xi|^2 q(\omega) + R, \quad q(\omega) = \langle \omega, U \rangle \langle \omega, N \rangle - \langle U, N \rangle,$$

with $|R| \leq (\vartheta^3/6 + \vartheta^4/24) |U| |N|$.

Proof. Rodrigues' formula $e^{[\xi]} = I + \sin \vartheta [\omega] + (1 - \cos \vartheta) [\omega]^2$, $[\omega] U = \omega \times U$, $\langle [\omega]^2 U, N \rangle = q(\omega)$, and $|\sin \vartheta - \vartheta| \leq \vartheta^3/6$, $|1 - \cos \vartheta - \vartheta^2/2| \leq \vartheta^4/24$, $\|[\omega]\| = \|[\omega]^2\| = 1$. $\square$

Lemma 4.4 (contact expansion). *Fix a contact $c = (i, j, \phi, \chi)$, $\hat{w}$ and ω , and let*

$$g_0 = U_i(0) \times N_j, \quad b(\hat{w}) = \frac{d}{ds} \Big|_{s=0} U_i(s\hat{w}) \times N_c, \quad \alpha_k(\hat{w}) = \frac{d}{ds} \Big|_{s=0} \langle \Pi(W_k - U_i)(s\hat{w}), n_c \rangle,$$

$f = |\hat{w}|_1 + |\phi| + |\chi| |\hat{w}_1|$ and $q_0 = q(\omega)$ computed with $U = U_i(0)$, $N = N_j$. Then for $k \in K_j$ and $\vartheta = |\xi|$,

$$\begin{aligned} \langle \Pi e^{[\xi]} U_i, n_c \rangle - \langle \Pi W_k, n_c \rangle &\geq \vartheta \left(\langle \omega, g_0 \rangle + s \langle \omega, b \rangle - \frac{1}{2} f^2 s^2 \right) + \frac{1}{2} \vartheta^2 q_0 - \frac{1}{2} \vartheta^2 f s - \left(\frac{\vartheta^3}{6} + \frac{\vartheta^4}{24} \right) \\ &\quad - s \alpha_k - \frac{1}{2} |d_{ik}| f^2 s^2. \end{aligned}$$

Proof. Write the left side as $\langle (e^{[\xi]} - I) U_i, N_c \rangle + \langle \Pi(U_i - W_k), n_c \rangle$ and apply Lemma 4.3 with $U = U_i(s\hat{w})$, $N = N_c(s)$, $|U| = |N| = 1$. Along $s \mapsto s\hat{w}$ the frame $A(s\hat{w})$ has derivatives of norm $\leq |\hat{w}|_1$ and second derivatives of norm $\leq |\hat{w}|_1^2$ (Lemma 2.2), and n_c rotates at speed $|\phi + \chi \hat{w}_1| \leq |\phi| + |\chi| |\hat{w}_1|$. Hence $|\frac{d^2}{ds^2} (U \times N)| \leq f^2$, which gives the first bracket. For q note $\frac{d}{ds} q = -\langle U', N - \omega \langle \omega, N \rangle \rangle - \langle N', U - \omega \langle \omega, U \rangle \rangle$, so $|\frac{d}{ds} q| \leq |U'| + |N'| \leq f$. Finally $s \mapsto \langle \Pi A(s\hat{w}) d_{ik}, n_c(s) \rangle$ vanishes at 0 (as $i \in I_j$, $k \in K_j$), has derivative α_k at 0 and second derivative bounded by $|d_{ik}| f^2$. $\square$

Lemma 4.5 (inactive vertices). *For a contact $c = (i, j, \phi, \chi)$ let*

$$\Delta_j = \min_{k \notin K_j} (H_j - \langle \Pi W_k(0), n_j \rangle) > 0$$

and let

$$a_k = \max \left(|\partial_{w_1} \langle \Pi(W_k - U_i), n_j \rangle| + |\partial_{w_2} \langle \Pi(W_k - U_i), n_j \rangle|, |\langle \Pi(W_k - U_i)(0), n_j^\perp \rangle| \right)$$

at $w = 0$, $a_{\text{in}} = \max_{k \notin K_j} a_k$, $a_{\text{act}} = \max_{k \in K_j} a_k$. If

$$r_0 (a_{\text{in}} + a_{\text{act}}) (1 + |\phi| + |\chi|) + 4r_0^2 (\sqrt{2} + |\phi| + |\chi|)^2 < \Delta_j,$$

then for $|\hat{w}|_\infty \leq 1$ and $0 < s \leq r_0$ the maximum of $\langle \Pi W_k(s\hat{w}), n_c \rangle$ over all k is attained in K_j .

Proof. For $k \notin K_j$, $k' \in K_j$ let $e(s) = \langle \Pi(W_k - W_{k'})(s\hat{w}), n_c(s) \rangle$, so $e(0) \leq -\Delta_j$. Writing $W_k - W_{k'} = (W_k - U_i) - (W_{k'} - U_i)$ gives $|e'(0)| \leq (a_k + a_{k'})(1 + |\phi| + |\chi|)$ (the derivative of the frame contributes at most $|\partial_{w_1} \cdot| + |\partial_{w_2} \cdot|$ since $|\hat{w}|_\infty \leq 1$, the rotation of n_c at speed $\leq |\phi| + |\chi|$ contributes the distance along the line), and $|e''| \leq |V_k - V_{k'}|(|\hat{w}|_1 + |\phi| + |\chi|)^2 \leq 2(2 + |\phi| + |\chi|)^2 \leq 8(\sqrt{2} + |\phi| + |\chi|)^2$. Hence $e(s) < 0$ for $0 < s \leq r_0$. $\square$

Contacts violating the hypothesis of Lemma 4.5 are discarded, so for the remaining contacts $P_c = \min_{k \in K_j} (\langle \Pi e^{[\xi]} U_i, n_c \rangle - \langle \Pi W_k, n_c \rangle)$.

4.5. Sector theorem. Divide the bound of Lemma 4.4 by $\vartheta = s\tau$ (ξ -mode) or by s (s -mode). Using $\vartheta = s\tau$, $\tau \leq 1$ and $s \leq r_0$, one obtains for every contact c a lower bound

$$(3) \quad \frac{P_c}{\vartheta} \text{ or } \frac{P_c}{s} \geq F_c(s) := \tau_* \langle \omega, g_0 \rangle + e_c(s) + s(\tau_* \langle \omega, b \rangle + \beta_c) - s^2 C_c,$$

where $\tau_* = 1$ in ξ -mode and $\tau_* = \tau$ in s -mode, and

$$\xi\text{-mode: } e_c(s) = -\frac{1}{\tau} \max_{k \in K_j} (\alpha_k + \frac{1}{2} |d_{ik}| f^2 s)^+, \quad \beta_c = \frac{1}{2} \tau q_0, \quad C_c = \frac{1}{2} f^2 + \frac{1}{2} \tau f + \frac{1}{6} \tau^2 (1 + \frac{r_0}{4}),$$

$$s\text{-mode: } e_c(s) = -\max_{k \in K_j} (\alpha_k + \frac{1}{2} |d_{ik}| f^2 s), \quad \beta_c = \frac{1}{2} \tau^2 q_0, \quad C_c = \frac{1}{2} \tau f^2 + \frac{1}{2} \tau^2 f + \frac{1}{6} \tau^3 (1 + \frac{r_0}{4}).$$

(In ξ -mode the negative part of α_k is dropped, which only weakens the bound.) Each e_c is concave in s . The ξ -mode is used only at A and B , where $T \in I$; there the configurations with $\xi = 0$ are coincidences with the reference and are not passages. In s -mode no division by $|\xi|$ occurs. For the supporting directions at vertices the normal n_j is a floating-point unit vector, $|n_j| \leq 1 + 10^{-15}$; all remainder constants are multiplied by $1 + 10^{-12}$ to account for this.

Theorem 4.6 (sector certificate). *Let Ω be a spherical cap with centre ω_c and radius ρ , $\mathcal{W} \subset \mathbb{R}^2$ a box, and $[\tau_l, \tau_h] \subset [0, 1]$; let $\mathcal{S}$ be the set of configurations with $0 < s \leq r_0$, $\omega \in \Omega$, $\hat{w} \in \mathcal{W}$, $\tau \in [\tau_l, \tau_h]$. Suppose there are weights $\lambda_c \geq 0$, not all zero, such that for $\tau \in \{\tau_l, \tau_h\}$ in s -mode, respectively for $\tau_* = 1$ in ξ -mode (where τ enters only through e_c, β_c, C_c , which are bounded using $\tau \in [\tau_l, \tau_h]$; if $\tau_l = 0$ in ξ -mode, assume in addition $\alpha_k + \frac{1}{2} r_0 |d_{ik}| f^2 \leq 0$ for all active partners of all contacts with $\lambda_c > 0$, so that $e_c \equiv 0$),*

$$(i) \quad \tau_* (\langle \omega_c, G \rangle - \rho |G|) + \sum_c \lambda_c \underline{e}_c(0) \geq 0,$$

$$(ii) \quad \tau_* (\langle \omega_c, H \rangle - \rho |H|) + \sum_c \lambda_c \left(\underline{e}_c(r_0) + r_0 (\underline{\beta}_c - \tau_* \|\partial b\| \text{rad}(\mathcal{W})) - r_0^2 \overline{C}_c \right) > 0,$$

where $G = \sum \lambda_c g_{0,c}$, $H = \sum \lambda_c (g_{0,c} + r_0 b_c(\hat{w}_c))$, $\hat{w}_c$ is the centre of $\mathcal{W}$, $\|\partial b\|$ the Frobenius norm of the linear map $\hat{w} \mapsto b(\hat{w}) - b(0)$, and underlines/overlines denote lower/upper bounds over $\mathcal{W} \times \Omega \times [\tau_l, \tau_h]$. Then $\mathcal{S}$ contains no passage.

Proof. Fix a configuration in $\mathcal{S}$. The function $F(s) = \sum_c \lambda_c F_c(s)$ is concave on $[0, r_0]$ (each e_c is concave and $-s^2 C_c$ is concave). Since $|\omega - \omega_c| \leq \rho$ we have $\langle \omega, G \rangle \geq \langle \omega_c, G \rangle - \rho |G|$, and $\langle \omega, b(\hat{w}) \rangle \geq \langle \omega, b(\hat{w}_c) \rangle - \|\partial b\| |\hat{w} - \hat{w}_c|$; the dependence on τ is affine in the terms containing $\langle \omega, \cdot \rangle$, and the remaining terms are bounded using the extreme values of τ . Thus (i), (ii) give $F(0) \geq 0$ and $F(r_0) > 0$, so by concavity $F(s) > 0$ for $0 < s \leq r_0$. By (3), $\sum_c \lambda_c P_c > 0$, so $P_c > 0$ for some c . $\square$

Remark 4.7. Applying the cap slack $\rho|\cdot|$ to the *combined* vector G rather than to each $g_{0,c}$ is essential: for rotations about in-plane axes parallel to an edge normal the first-order terms of two contacts cancel exactly.

Lemma 4.8 (exact cancellation). *Let p, q be contacts whose inner vertices lie in the shadow plane, $\langle U_{i_p}(0), e_3 \rangle = \langle U_{i_q}(0), e_3 \rangle = 0$, so that $g_{0,p} = (0, 0, a)$ and $g_{0,q} = (0, 0, -b)$, with $a, b > 0$, and suppose $\alpha_k \leq 0$ for all active partners of p and q on $\mathcal{W}$ (in ξ -mode with $\tau_l = 0$, assume the stronger hypothesis of Theorem 4.6). Then with $\lambda = (b, a)/(a + b)$ we have $G = 0$, and condition (i) holds since $e_c(0) \geq 0$ for both contacts (in ξ -mode $e_c(0) = 0$); condition (ii) can be verified with interval enclosures of λ .*

The hypothesis $\langle U_i(0), e_3 \rangle = 0$ is verified in $\mathbb{Q}(\sqrt{5})$. Partners with identical trajectories ($d_{ik} = 0$, verified in $\mathbb{Q}(\sqrt{5})$) have $\alpha_k \equiv 0$ exactly.

Finding the weights. The weights are computed by a linear program (HiGHS) that maximises the slack of (ii) subject to (i); they are only witnesses, and (i), (ii) are then verified in interval arithmetic. Because of the solver's feasibility tolerance ($\approx 10^{-7}$) the returned weights occasionally violate (i) by a tiny amount (we observed $\approx -2 \cdot 10^{-9}$) near configurations where the first-order term of the optimal combination is close to zero. The program therefore also tries: (a) Lemma 4.8 applied to an opposite pair of exact-zero contacts contained in the support of the LP solution (the LP sometimes adds a very small weight, $\approx 10^{-5}$, on a third contact); (b) re-solving the LP with (i) required with a margin $10^{-8}, 10^{-7}, 10^{-6}$. These are strategies for finding witnesses only; every accepted box satisfies the hypotheses of Theorem 4.6 or Lemma 4.8, verified rigorously.

4.6. Arc-tube theorem. At C and D the protrusion vanishes along the half-twist arc $\{w = (0, \pm h), \xi = 0\}$, so no sector certificate can be valid near the ray $\hat{w} = (0, \pm 1)$, $\tau = 0$. We therefore use the normalisation $N = |\xi| + |u|$, where $w = (u, \pm h)$.

Theorem 4.9 (arc-tube certificate). *Let $\varepsilon_0, \beta, r_0 > 0$ and $\mathcal{T} = \{w = (u, \pm h) : 0 < h \leq r_0, |u| \leq \varepsilon_0 h, |\xi| \leq \beta h\}$. Contacts are $c = (i, j, \phi, \chi)$ with $n_c = R(h\phi + u\chi)n_j$. An active partner $k \in K_j$ is exact if $e_k(h, u) := \langle \Pi(W_k - U_i)(u, \pm h), n_c \rangle$ vanishes identically for $u = 0$ (identical trajectories $d_{ik} = 0$, or collinearity persisting along the arc; both verified in $\mathbb{Q}(\sqrt{5})$). With $\tau = |\xi|/N$, $\sigma = \text{sign } u$,*

$$\frac{P_c}{N} \geq \Phi_c + h\Psi_c - h^2 K_c,$$

$$\Phi_c = \tau \langle \omega, g_0 \rangle - (1 - \tau) D_\sigma, \quad D_\sigma = \max_{k \text{ exact}} \sigma(t_k + \chi p_k),$$

$$\Psi_c = \tau \langle \omega, g_1 \rangle - \tau \varepsilon_0 (1 + |\chi|) + \frac{1}{2} \tau \beta \min(q_0, 0) - (1 - \tau) (M_X + \frac{1}{2} \varepsilon_0 M_{TT}),$$

$$K_c = \frac{1}{2} f^2 + \beta (1 + \varepsilon_0) (1 + |\phi| + |\chi|) + \frac{1}{6} \beta^2 (1 + \frac{\beta r_0}{4}), \quad f = 1 + \varepsilon_0 + |\phi| + \varepsilon_0 |\chi|,$$

where $g_1 = \partial_h(U_i \times N_c)$ at 0, $M_X = \max_k |d_{ik}|(1 + |\phi|)(1 + |\chi|)$ and $M_{TT} = \max_k |d_{ik}|(1 + |\chi|)^2$ over exact partners, provided every outer vertex k which is not an exact partner satisfies, with $\Delta_k = \frac{1}{2} r_0 |d_{ik}| f^2 + \varepsilon_0 (|t_k| + |\chi| |p_k|) + \varepsilon_0 (D_{\max} + r_0 (M_X + \frac{1}{2} \varepsilon_0 M_{TT}))$,

$$g_k + r_0 (|\alpha_k| + \Delta_k) < 0 \quad \text{if } g_k < 0, \quad \alpha_k + \Delta_k < 0 \quad \text{if } g_k = 0$$

(with $g_k = e_k(0, 0) \leq 0$, $\alpha_k = \partial_h e_k(0, 0)$ with its sign, $t_k = \langle \Pi \partial_{w_1}(W_k - U_i)(0), n_j \rangle$ the frame part and $p_k = \langle \Pi(W_k - U_i)(0), n_j^\perp \rangle$, so that $\partial_u e_k(0, 0) = t_k + \chi p_k$, and $D_{\max} = \max |t_k + \chi p_k|$ over exact

partners). Exactness by collinearity along the arc is used only for contacts with $\phi = 0$, for which the line normal n_c does not depend on h when $u = 0$. Consequently, if weights $\lambda \geq 0$, not all zero, satisfy $\sum \lambda_c \Phi_c \geq 0$ and $\sum \lambda_c (\Phi_c + r_0 \Psi_c - r_0^2 K_c) > 0$ on a box of (ω, τ) and a sign σ (with the cap slack applied to the combined vectors; the ω -terms are affine in τ , so the conditions are checked at the endpoints of each τ -interval), that part of $\mathcal{T}$ contains no passage. An exact-zero variant of Lemma 4.8 is also used: if the support is an opposite pair of contacts with $g_0 = (0, 0, \pm)$ (claims Z) and $D_\sigma = 0$ exactly (an exact partner with identical trajectory, $d_{ik} = 0$, and $\sigma(t_k + \chi p_k) < 0$ for all other exact partners), then the weights $(b, a)/(a + b)$ give $\sum \lambda_c \Phi_c = 0$ exactly. Configurations with $N = 0$ ($u = 0$, $\xi = 0$) lie on the half-twist arc; there the inner vertex of a contact lies on the line of an exact partner (exact claims, Section 6.3), which is a supporting line of the outer shadow by the separation condition for the non-exact vertices at $u = 0$; so these configurations are not passages either.

Proof. Exact partners. For fixed h , $e_k(h, 0) = 0$, so Taylor's theorem in u gives $e_k(h, u) = u \partial_u e_k(h, 0) + \frac{1}{2} u^2 \partial_u^2 e_k(h, u')$ and $\partial_u e_k(h, 0) = \partial_u e_k(0, 0) + h \partial_h \partial_u e_k(h', 0)$. The mixed and second derivatives are bounded by M_X and M_{TT} (Lemma 2.2), hence, with $|u| \leq \varepsilon_0 h$, $e_k \leq |u|(\sigma \partial_u e_k(0, 0) + h(M_X + \frac{1}{2} \varepsilon_0 M_{TT}))$ and $|u| = (1 - \tau)N$. *Non-exact vertices.* By the displayed condition (a Taylor bound in (h, u) for e_k : $e_k(h, u) \leq g_k + h\alpha_k + \frac{1}{2} h^2 |d_{ik}| f^2 + |u|(|t_k| + |\chi| |p_k|)$, and the bound $|e_{k'}| \leq \varepsilon_0 h(D_{\max} + r_0(\dots))$ for exact partners k' ; if $g_k < 0$ we bound $h \leq r_0$, if $g_k = 0$ we divide by h) every outer vertex which is not an exact partner lies strictly below the exact partners throughout $\mathcal{T}$, so the maximum in P_c is attained at an exact partner. *Inner term.* By Lemma 4.3, $\langle (e^{[\xi]} - I)U_i, N_c \rangle \geq \langle \xi, U_i \times N_c \rangle + \frac{1}{2} |\xi|^2 q - |\xi|^3 (1 + |\xi|/4)/6$. Along the path $t \mapsto (tu/h, \pm t)$, $0 \leq t \leq h$, the vector $U_i \times N_c$ has derivative of norm $\leq f$ and second derivative $\leq f^2$, and $|\partial_u (U_i \times N_c)| \leq 1 + |\chi|$; hence $\langle \xi, U_i \times N_c \rangle \geq |\xi|(\langle \omega, g_0 \rangle + h \langle \omega, g_1 \rangle - |u|(1 + |\chi|) - \frac{1}{2} f^2 h^2)$. Moreover $|\xi| \leq \beta h$, $|q - q_0| \leq 2(1 + \varepsilon_0)(1 + |\phi| + |\chi|)h$ and $|\xi|/N = \tau \leq 1$. Dividing by N and collecting terms gives the stated bound; it is concave in h , so the two conditions imply $P_c > 0$ for some c and all $0 < h \leq r_0$. $\square$

4.7. Covering a neighbourhood of a singular point. At A and B we cover $\{s \leq r_0\}$ by two pieces, $|\xi| = s \geq |w|$ (piece 1, $\tau = 1$, $\hat{w}$ in the unit square) and $|w| = s \geq |\xi|$ (piece 2, $\hat{w}$ on the unit circle, $\tau \in [0, 1]$), each subdivided along the six faces of the cube for ω . At A the reflection coincidence passes through x^* with $\xi = \psi(w) = \log(\rho_\pi A(w) h A(w)^T) = D\psi w + O(|w|^2)$, $D\psi = \begin{pmatrix} 0 & 2 \\ 2 \sin \phi^* & 0 \\ 0 & 0 \end{pmatrix}$ ($2 \sin \phi^* = 1.0515\dots$); boxes with $|\tau\omega - D\psi\hat{w}| \leq 0.6|\hat{w}|$ (after all slacks, including $0.02|\hat{w}|$ for the nonlinearity of ψ) are covered by the sector certificate for the second reference $\rho_\pi A(w)h$ (piece 2 with $\tau \leq 0.8$). The precise statement is Lemma A.3 in the appendix. At C and D the s -mode sector certificate covers $\{s \leq r_0\}$ except the arc tubes, which are covered by Theorem 4.9 (at C two tubes: the half-twist reference and the half-twist composed with the reflection); see Lemma A.4.

5. PROOF OF THEOREM 1.1

By Lemma 2.1 it suffices to show that $\mathcal{D}$ contains no passage. For each singular point $X \in \{A, B, C, D\}$ with radius r_X ($r_A = r_B = 1.5 \cdot 10^{-3}$, $r_C = r_D = 10^{-3}$) let $\mathcal{B}_X$ be the set of configurations whose outer view is within Euclidean distance r_X of (θ^*, ϕ^*) in the (θ, ϕ) -plane and whose inner orientation satisfies $A_{\text{in}} = e^{[\xi]} D A(w) T_X g$ with $|\xi| \leq r_X$ for some $g \in I$, $D \in \{I, \rho_\pi\}$.

- (1) *Balls.* By Lemma A.1, every configuration in $\mathcal{B}_X$ has the same shadows as a configuration (w, ξ') in blow-up coordinates of the reference $A(w) T_X$ with $s = \max(|w|, |\xi'|) \leq r_X$. The set $\{0 < s \leq r_X\}$ is the union of piece 1 ($|\xi| = s \geq |w|$) and piece 2 ($|w| = s \geq |\xi|$), and each piece is covered by finitely many regions $\mathcal{S}$ of Theorem 4.6, except for regions covered by Theorem 4.6 for a second reference (at A , Lemma A.3) or by Theorem 4.9 (at C, D ,

Lemma A.4). At $s = 0$ the configuration is the singular configuration itself: at A, B a coincidence, at C, D an endpoint of the half-twist arc, where an inner vertex lies on a line j of the outer shadow (exact claims), which is a supporting line since $\Delta_j > 0$ (decided in decimal arithmetic, Appendix C); in both cases it is not a passage.

- (2) *Cells*. Every cell of $\mathcal{D}$ is subdivided recursively; each leaf box is closed by [8, Theorem 17], [8, Theorem 36], Theorem 4.1 or Theorem 4.2, or is contained in some $\mathcal{B}_X$. Containment is certified by bounding the view distance and the rotation angle (Lemma 2.2: the angle changes by at most 5ε over a box of half-width ε).

All hypotheses are verified by the computation described in Section 6. □

6. COMPUTATION

6.1. Arithmetic. Interval arithmetic uses IEEE double precision with outward rounding by one unit in the last place after every operation (`nextafter`); sums or products with an exact zero interval are exact. Sines and cosines are enclosed without relying on the platform library: the argument is reduced by $k\pi/2$ with $\pi \in [\text{fl}(\pi), \text{fl}(\pi)^+]$ in interval arithmetic, and $\sin r, \cos r$ ($|r| < 0.8$) are evaluated by their Taylor polynomials of degree 21 and 20 in interval arithmetic with the Lagrange remainders $|r|^{23}/23!$, $|r|^{22}/22!$; interior extrema are treated explicitly. The remainder bounds are evaluated with the library power function; they are below 10^{-7} units in the last place of the result, so any rounding error in them is absorbed by the final outward rounding. (We compared the enclosures with 70-digit values at 6000 points as a test of the implementation.) Vertex coordinates in the rotated frame are decimal values with error below 10^{-55} , enclosed as described in Appendix C. Exact identities are verified with a small implementation of $\mathbb{Q}(\sqrt{5})$ with rational coefficients.

6.2. Independent checks. A second implementation of Theorems 4.6, 4.9 and Lemma 4.8, written from the statements in this paper and using exact rational interval arithmetic, takes from the main programs only the combinatorial data of the contacts (inner vertex, line normal, parameters, active sets) and the certificate accepted by the main program (support, enclosed weights, and whether Lemma 4.8 was used; in the latter case the sector checker verifies that the enclosures of $g_{0,p}, g_{0,q}$ are consistent with $(0, 0, a)$, $(0, 0, -b)$ and that the weights enclose $(b, a)/(a + b)$, while the tube checker only verifies $D_\sigma \leq 0$ and takes the exact cancellation from the main program). All numerical quantities are recomputed from the decimal data. With the final version of the programs it re-verified 3500 sector boxes, namely 3000 random boxes (2×300 for each of A with both references, B, C, D) and 500 boxes from the regions where the witness search needed the strategies (a), (b) described after Lemma 4.8 (300 near A , all certified by Lemma 4.8, and 200 near C), and 500 tube boxes; all were confirmed. For the global theorem, a separate program subdivides 3000 randomly chosen cells by itself (all 32 children, depth at most 6) and re-verifies up to three Global-Theorem leaves per cell (7290 leaves) in exact rational interval arithmetic (vertices enclosed with radius 10^{-50} , sines and cosines by Taylor series with explicit remainder); all were confirmed.

These checks have a limited scope. The sampled sector boxes are random dyadic boxes of depth 6 to 11 (tube boxes: depth 3 to 8) which are certified at sampling time, not the leaves of the actual cover (which reaches depth 28); ξ -mode boxes with $\tau_l = 0$ are not sampled; and both implementations use the same exact data of the polyhedron. In an earlier version of the main program the second implementation detected an omitted term in the inactive-vertex condition (Lemma 4.5), which was corrected before the runs reported here.

A third implementation (`indep_leaves.py`), written from the statements in this paper and in [8], re-checks *actual leaves* of the cell cover that are closed by the other certificates. It shares no code with the main programs (it uses them only to read the numbering of the vertices): it rebuilds the vertices from the definition in Section 2 with 80-digit decimals, proves the vertex permutations of I and the congruences exactly in $\mathbb{Q}(\sqrt{5})$, and checks the hypotheses of [8, Theorem 36] as stated there and of Theorems 4.1 and 4.2, and ball membership, in exact rational interval arithmetic. For

Theorem 4.1 it chooses its own contact directions, and it bounds the rotation angle through the identity $\|Q - I\|_F^2 = \frac{1}{20} \sum_k |(Q - I)V_k|^2$ (the vertices form a tight frame, verified exactly); only the lower bound for c_0 uses floating-point dot products, with an explicit rounding margin. The leaves were obtained by re-running the prover on randomly chosen cells (the regenerated covers agreed with the recorded leaf counts in every cell): 630 SY-local leaves from 59 cells, 640 leaves of Theorem 4.1 from 65 cells, 925 leaves of Theorem 4.2 from 58 cells, and 378 ball-membership leaves from 3 cells; all were confirmed. (With five contact directions per vertex of the outer shadow, two leaves of Theorem 4.1 were not confirmed, because c_0 depends on this choice; with eleven directions all were.) In the same way, 1332 actual leaves of the Global Theorem (the smallest ones of 40 cells, half of them cells near the singular points, down to relative depth 10) were re-checked with the exact rational implementation above; all were confirmed.

As tests of the implementation (not part of the proof), the per-contact lower bounds of Theorem 4.6 were compared with the exact protrusion computed directly from the geometry at $3.5 \cdot 10^5$ random (contact, configuration) pairs with $s = r_0$ near A, B, C, D , and those of Theorem 4.9 at $8.3 \cdot 10^5$ pairs with $h = r_0$; the bound was never violated (smallest slack $2 \cdot 10^{-8}$).

6.3. Exact identities. All structural statements used by the certificates are reduced to polynomial identities between exact vectors in $\mathbb{Q}(\sqrt{5})^3$ (body frame): the vertices V_k , the singular views X (Proposition 3.1), the five-fold axis $w_5 = (0, \varphi, 1)$, the half-twists T and the group elements. With p an inner vertex ($p = \sigma TV_i$) and $e = V_{q'} - V_q$ the direction of an outer edge, they are:

- *on a shadow line at x^** : $\det[p - V_q, e, X] = 0$;
- *coincident shadow points*: $(p - V_v) \times X = 0$;
- *identical trajectories*: $V_k = p$;
- *in the shadow plane*: $\langle p, X \rangle = 0$;
- *exact along the arc*: with $m_1 = w_5 \times X$, $m_2 = X \times m_1$ and $d = V_k - p$, $\langle m_1, e \rangle \langle X, d \rangle = \langle m_1, e \rangle \langle m_2, d \rangle = \langle m_2, e \rangle \langle m_1, d \rangle = 0$.

For the last item: along the arc the view $X(h)$ stays in the mirror plane $\Pi_0 = m_1^\perp$, the first image axis is $\pm m_1/|m_1|$ and the second is $m_2(h) \in \Pi_0$; the fixed line normal is $n \propto (-\langle m_2, e \rangle, \langle m_1, e \rangle)$ at $h = 0$. Then $e_k(h) \propto -\langle m_2, e \rangle \langle m_1, d \rangle + \langle m_1, e \rangle \langle m_2(h), d \rangle$. If $\langle m_1, e \rangle \neq 0$ the first two identities say that the component of d in Π_0 vanishes, so $\langle m_2(h), d \rangle \equiv 0$, and the third gives $e_k \equiv 0$; if $\langle m_1, e \rangle = 0$, e_k is constant in h and vanishes by the third identity; this argument uses the fixed normal, which is why it is applied only to contacts with $\phi = 0$. The sector certificates use 606 such claims and the three tube certificates 1048, 1048 and 1052; none fails. In addition, the Gram-matrix equalities for the local theorem of [8] are verified in $\mathbb{Q}(\sqrt{5})$ during the cell cover.

6.4. Reproducibility. The programs (Python, with `numpy` and `scipy`; linear programs are solved with HiGHS only to produce witnesses) are listed in the repository with instructions. On a laptop with six cores the tubes take about one hour each, the sectors and the cell cover together a few days of CPU time. All runs are resumable and write their results to text files; a run succeeds when it reports no failure.

6.5. Organisation.

- (1) The domain $\mathcal{D}$ is divided into $12 \times 7 \times 12 \times 7 \times 30 = 211,680$ cells. The end points $2\pi/5$, $\phi_{\max}$ and $\pm\pi/2$ are replaced by their outward floating-point roundings, so that the union of the cells contains $\mathcal{D}$; adjacent cells and boxes share their floating-point boundaries. Each cell is subdivided recursively and every leaf is closed by an interval-verified certificate (global, SY-local, Theorem 4.1 or 4.2), or lies in a certified ball around a singular point.
- (2) Around A, B ($r_0 = 1.5 \cdot 10^{-3}$) and C, D ($r_0 = 10^{-3}$) the sector and tube certificates cover $\{s \leq r_0\}$. A box belongs to such a ball if its outer view is within r_0 of (θ^*, ϕ^*) and its relative rotation to some reference $DA(w)T_X g$ ($g \in I$) has angle at most r_0 ; all references

$DA(w)T_X g$ give the same inner shadow (Lemma A.1), so the ball is covered by the certificate for one reference.

part	size	result
cell cover, all 211,680 cells	$1.23 \cdot 10^7$ leaves	0 failures
exact structural claims in $\mathbb{Q}(\sqrt{5})$ (sectors, tubes)	606 + 3148	0 violations
arc tubes C (two references), D	$2 \times 6963 + 6662$ boxes	0 failures
sectors A (two references), B, C, D	$3.61 \cdot 10^6$ boxes	0 failures
independent rational re-check: sectors / tubes	3500 / 500 boxes	all confirmed
independent rational re-check: global leaves	7290 leaves	all confirmed
third implementation, actual leaves:		
SY-local / Thm 4.1 / Thm 4.2 / ball	630 / 640 / 925 / 378	all confirmed
global (deepest leaves of 40 cells)	1332	all confirmed

TABLE 2. Summary of the computation. All runs use the final version of the rigorous checks; during the sector runs only the witness search (Section 4) and the maximal subdivision depth were extended, which affects which boxes are subdivided but not the verification of a box. Leaves of the cell cover: 11,439,400 global, 448,173 SY-local, 46,102 Theorem 4.1, 288,772 Theorem 4.2, 33,880 in certified balls (about 38 CPU hours). Sector boxes: $1.51 \cdot 10^6$ (A), $1.23 \cdot 10^6$ (B), $8.2 \cdot 10^5$ (C), $5.6 \cdot 10^4$ (D); these counts include empty boxes of piece 1 outside the unit disc. In addition, 83,538 (A), 27,668 (C) and 24,792 (D) boxes were assigned to another certificate by the skip lemmas of Appendix A.

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APPENDIX A. COVERING LEMMAS

Lemma A.1 (references are interchangeable). *Let T be a fixed rotation, $g \in I$ and $D \in \{I, \rho_\pi\}$. If $A_{\text{in}} = e^{[\xi]} DA(w)Tg$, then the inner shadow equals that of $e^{[\xi']} A(w)T$ with $\xi' = D^T \xi$, $|\xi'| = |\xi|$.*

Proof. $gV = V$ as sets. For $D = \rho_\pi$: $e^{[\xi]} \rho_\pi = \rho_\pi e^{[\rho_\pi^T \xi]}$ and $\Pi \rho_\pi = -\Pi$, so the shadow is $-\Pi e^{[\xi']} A(w)TV = \Pi e^{[\xi']} A(w)T(-V)$, and $-V = V$. $\square$

Lemma A.2. *Let $\Phi(w) = \rho_\pi A(w)MA(w)^T$ with M a rotation such that $\Phi(0) = I$ (at A , $M = h$ is the half-turn about X_A ; at C , $M = T_C h_2 T_C^T$ is the half-turn about X_C ; both facts are verified exactly in $\mathbb{Q}(\sqrt{5})$ by `check_branches_exact.py`), and $\psi(w) = \log \Phi(w)$ (rotation vector). For $|w|_1 \leq 10^{-2}$, $|\psi(w) - D\psi w| \leq 3|w|_1^2$.*

Proof. By Lemma 2.2, second derivatives of $w \mapsto A(w)MA(w)^T$ are sums of at most four products with norm ≤ 1 , so $\|\Phi(w) - I - \Phi'(0)w\| \leq 2|w|_1^2$ and $\|\Phi(w) - I\| \leq 2|w|_1 + 2|w|_1^2$. For a rotation $Q = e^{[\eta]}$ the skew part satisfies $\frac{1}{2}(Q - Q^T) = \sin|\eta| [\hat{\eta}]$, hence $|\eta - \text{vee}(\frac{1}{2}(Q - Q^T))| \leq |\eta|^3/6$ and $|\eta| \leq 1.01\|Q - I\|$ for small $|\eta|$. Since $\Phi'(0)w$ is skew and equals $[D\psi w]$, the skew part gives $|\text{vee}(\frac{1}{2}(\Phi - \Phi^T)) - D\psi w| \leq \|\Phi - I - \Phi'(0)w\| \leq 2|w|_1^2$. With $\eta = \psi(w)$, $|\eta| \leq 1.01 \cdot (2|w|_1 + 2|w|_1^2) \leq 0.021$, hence $|\eta|^3/6 \leq 1.5|w|_1^3 \leq 0.015|w|_1^2$, and altogether $|\psi(w) - D\psi w| \leq 2.02|w|_1^2 \leq 3|w|_1^2$. $\square$

Numerically the constant is 0.45 at A and 0.09 at C .

Lemma A.3 (skip region at A). *Let x be a configuration with $s \leq r_0 = 1.5 \cdot 10^{-3}$ in the blow-up coordinates of the identity reference at A , with $|\xi - D\psi w| \leq 0.6|w| - 0.02|w|$. Then in the coordinates of the reflection reference $\rho_\pi A(w)h$ one has $|\xi_2| \leq 0.8|w|$, so x lies in piece 2 of the sector for that reference with $\tau \leq 0.8$.*

Proof. $e^{[\xi]} = e^{[\xi_2]}e^{[\psi(w)]}$, so $|\xi_2| = d(e^{[\xi]}, e^{[\psi(w)]})$ in the bi-invariant metric of $SO(3)$. Since $SO(3)$ has non-negative sectional curvature, $\exp$ is 1-Lipschitz on the ball of radius π (Rauch comparison), hence $|\xi_2| \leq |\xi - \psi(w)| \leq |\xi - D\psi w| + 3|w|_1^2 \leq 0.58|w| + 6r_0|w| \leq 0.8|w|$. $\square$

Lemma A.4 (skip regions at C). *In the s -mode sector at C (piece $|w| = s$), write $\hat{w} = (\cos \vartheta, \sin \vartheta)$. If $|\vartheta - \pi/2| \leq 0.378$ and $\tau \leq 0.46$ then $w = (u, h)$ with $|u| \leq \tan(0.378)h < 0.4h$ and $|\xi| \leq 0.46h/\cos(0.378) < 0.5h$, i.e. x lies in the tube of Theorem 4.9 with $\varepsilon_0 = 0.4$, $\beta = 0.5$. At D the same holds with ϑ near $-\pi/2$ and $w = (u, -h)$ (only this tube occurs at D).*

At C there is a second branch: $T_C h_2 T_C^T$ is the half-turn about X_C for a suitable $h_2 \in I$, so the second reference $\rho_\pi A(w) T_C h_2$ has $\Phi(w) = \rho_\pi A(w) T_C h_2 T_C^T A(w)^T$ with $\Phi(0) = I$, and $\psi(w) = \log \Phi(w) = D\psi_C w + O(|w|^2)$ with $D\psi_C = \begin{pmatrix} 0 & 2 \\ 2 \sin \phi_C^ & 0 \\ 0 & 0 \end{pmatrix}$, $2 \sin \phi_C^* = 0.1796 \dots$. In both pieces of the s -mode sector at C : if $w = (u, h)$ with $h > 0$, $|u| \leq 0.39h$ and $|\xi - D\psi_C w| \leq 0.49h - 0.02|w|$, then x lies in the tube of Theorem 4.9 for the second reference with $\varepsilon_0 = 0.4$, $\beta = 0.5$.*

Proof. The first part is elementary. For the second, as in Lemma A.3 the relative rotation ξ_2 to the second reference satisfies $|\xi_2| \leq |\xi - \psi(w)| \leq |\xi - D\psi_C w| + 3|w|_1^2 \leq 0.49h - 0.02|w| + 6r_0|w| < 0.5h$, and the outer view, hence (u, h) , is the same for both references. $\square$

APPENDIX B. CONDITIONS CHECKED FOR THE LOCAL THEOREM

For a box with centre $(\theta_1, \phi_1, \theta_2, \phi_2, \alpha)$ and largest half-width ε (rounded upward) the program `rig_sylocal.py` follows the published implementation of [8, Theorem 36] by its authors. For inner vertices P_1, P_2, P_3 and outer vertices Q_1, Q_2, Q_3 proposed by the floating-point oracle it verifies in interval arithmetic (with $\rho = 1$): the triples are congruent (by a symmetry of $\mathcal{P}$ or by exact equality of Gram matrices); $s_P \langle P_k, X_1 \rangle > \sqrt{2}\varepsilon$ and $s_Q \langle Q_k, X_2 \rangle > \sqrt{2}\varepsilon$ for fixed signs s_P, s_Q ; the ε -spanning conditions $\langle R(\pi/2)M_1 P_u, M_1 P_v \rangle > 2\varepsilon(\sqrt{2} + \varepsilon)$ and likewise for the Q 's for $(u, v) \in \{(1, 2), (2, 3), (3, 1)\}$ [8, Definition 27]; $|M_2 Q_k| > r + 1.42\varepsilon$; and, with $\delta = \max_k \frac{1}{2}|R(\alpha)M_1 P_k - M_2 Q_k|$, for every outer vertex $A \neq Q_k$ the inequality

$$\begin{aligned} 2r(|M_2 Q_k|^2 - \langle M_2 A, M_2 Q_k \rangle - |A - Q_k|(2\sqrt{2}\varepsilon + 2\varepsilon^2)) \\ \geq (4.5\varepsilon + 2\delta)(|M_2 Q_k| + 1.42\varepsilon)(|M_2 A - M_2 Q_k| + 2.84\varepsilon), \end{aligned}$$

where $|A - Q_k|$ is the distance in $\mathbb{R}^3$. Here $M_i = M(\theta_i, \phi_i)$, $X_i = X(\theta_i, \phi_i)$ and r is the radius supplied by the oracle.

These checks imply the hypotheses of [8, Theorem 36] on the box $[\theta_1 \pm \varepsilon, \dots, \alpha \pm \varepsilon]$, which contains the given box. The vertices of $\mathcal{P}$ have norm exactly 1, so $\rho = 1$, and all enclosures contain the exact vertices. Congruence [8, Definition 34]: a symmetry g with $Q_k = gP_k$ gives $P_k = g^{-1}Q_k$; equal Gram matrices give an orthogonal L with $P_k = LQ_k$ (the linear isometry between the spans extends to $\mathbb{R}^3$; cf. [8, Lemma 35] and the remark after it). The two sign conditions are (A_ε) with $(-1)^{\sigma_P} = s_P$, $(-1)^{\sigma_Q} = s_Q$. The ε -spanning conditions are literally those of [8, Definition 27], since $\langle R(\pi/2)u, v \rangle = u_1 v_2 - u_2 v_1$. Because $1.42 > \sqrt{2}$, the radius check implies $\min_k |M_2 Q_k| > r + \sqrt{2}\varepsilon$. The value δ is enclosed, and its upper end is used, so $\delta \geq \max_k \frac{1}{2}|R(\alpha)M_1 P_k - M_2 Q_k|$. Finally, the displayed inequality, with a right-hand side that is positive, forces the numerator of (B_ε) to be positive. Dividing by $2r$ times the product of the denominators, and using $1.42 > \sqrt{2}$, $2.84 > 2\sqrt{2}$ (larger denominators) and $2.25 > \sqrt{5}$, it gives

$$\frac{\langle M_2 Q_k, M_2(Q_k - A) \rangle - 2\varepsilon|Q_k - A|(\sqrt{2} + \varepsilon)}{(|M_2 Q_k| + \sqrt{2}\varepsilon)(|M_2(Q_k - A)| + 2\sqrt{2}\varepsilon)} \geq \frac{2.25\varepsilon + \delta}{r} > \frac{\sqrt{5}\varepsilon + \delta}{r},$$

which is (B_ε) for $Q_i = Q_k$ and every other vertex $Q_j = A$. We compared these conditions line by line with the statement of [8, Theorem 36] (arXiv version 2).

APPENDIX C. FLOATING-POINT DECISIONS

Decimal data. The exact data (vertices, singular views, half-twists and group elements in the rotated frame) are computed as decimal numbers with at least 60 correct digits. A decimal value d approximating an exact number e is enclosed by $[-10^{-40}, 10^{-40}]$ if $|d| < 10^{-40}$ and by the two floating-point neighbours of $\text{fl}(d)$ otherwise. Both enclosures are valid because every enclosed value satisfies either $|d| \leq 3 \cdot 10^{-59}$ (these are the structural zeros, certified separately in $\mathbb{Q}(\sqrt{5})$) or $|d| \geq 0.009$, while the error of the decimal computations is below 10^{-55} ; the script `check_decimal_range.py` in the repository checks this range. The error bound for the input data is certified directly (`certify_exact_data.py`): for each of the 772 decimal input values (rotation, vertices, singular views, half-twists, h_2 , and the 60 group matrices) the exact value is enclosed, from its definition, by an interval with rational end points (square roots by integer square roots, $\cos 36^\circ = \varphi/2$, $\sin 36^\circ = \sqrt{10 - 2\sqrt{5}}/4$), and the decimal value lies within $2.94 \cdot 10^{-59}$ of every point of this interval. Derived decimal quantities (frames at the singular views, projected vertices and their derivatives, contact data, products such as $T_X g$) are computed from these inputs with 70-digit arithmetic in a few hundred operations on numbers of modulus $O(1)$; their error is many orders of magnitude below the thresholds (10^{-12} , 10^{-13} , 10^{-40}) at which decisions are taken. As a check, the whole preparation was repeated with 160-digit arithmetic (`verify_decimal_precision.py`): every decision derived from the data in the sector and tube certificates (contact rows, active sets, exact partners, separation decisions, the values D_σ , the number of exact claims) is identical.

The exact vertex coordinates in the rotated frame that are structurally zero are exactly 0 in the decimal data; the programs for the global and local theorems enclose all coordinates by the floating-point neighbours of their value, which for these coordinates gives an interval containing 0.

Decisions with slack. The following choices are made in floating point; each is a strict inequality whose slack exceeds the floating-point error of the few operations involved (below 10^{-14} on quantities of size $O(1)$) by many orders of magnitude, or is protected by an explicit factor:

- the inactive-vertex quantities a_k of Lemma 4.5 (multiplied by 1.01);
- the skip tests of Lemmas A.3, A.4 (the tested inequalities imply the required ones with relative margin at least $7 \cdot 10^{-3}$, and an additional 10^{-9} is added);
- the outer-view distance test for membership in a ball $\mathcal{B}_X$ (margin 10^{-12});
- the conversion $\vartheta \leq 2y/\sqrt{1-y^2}$ in Theorem 4.1 (multiplied by $1 + 10^{-12}$);
- the norm $\|\partial b\|$ in Theorem 4.6, computed as the upward-rounded square root of a floating-point sum of six upward-rounded squares (the sum itself is not rounded upward); it depends only on the contact, and for all 36,974 contact rows of the sector certificates the computed value was checked in exact rational arithmetic to be an upper bound (`verify_gwn.py`);
- the lower bound for c_0 in Theorem 4.1: the products of the (floating-point) cap centres with the interval bounds of the g_c are summed in floating point, and an absolute margin $3 \cdot 10^{-15}$ is subtracted (the rounding error of these sums of three products of numbers of modulus at most 1 is below 10^{-15}).

The directions n used in Theorem 4.1 and in the sector certificates are floating-point vectors with $|n| \leq 1 + 10^{-15}$; all terms that scale with $|n|$ are multiplied by $1 + 10^{-12}$.

Decisions in decimal arithmetic. The membership in the active sets K_j , I_j (a value with $|d| < 10^{-40}$) is confirmed by exact claims in $\mathbb{Q}(\sqrt{5})$; the complementary strict inequalities (e.g. $\Delta_j > 0$) and the separation conditions for non-exact vertices in Theorem 4.9 (thresholds 10^{-12} , 10^{-13}) are decided in decimal arithmetic, whose error (below 10^{-50}) is far below these thresholds. (For the tube conditions we checked that the decimal decisions agree with an earlier floating-point evaluation for all contacts.)

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INDEPENDENT RESEARCHER, REPUBLIC OF KOREA

Email address: swiri0921@gmail.com